

AWS SUMMIT



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Accelerate Generative AI: Amazon SageMaker HyperPod Training Plans & Recipes

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Agenda

- Business challenges: FM Training & Inference
- Introduction to Amazon SageMaker HyperPod
- Features
- Customer stories
- Demo

Business challenges: FM Training and Inference

FM training in practice

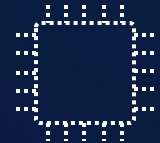
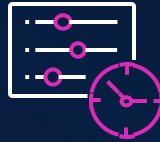
Pretraining Meta's Llama 3 405B

- 54 days of continuous training
- 16,000 NVIDIA H100 Tensor Core GPUs
- 419 unexpected interruptions
 - 78% attributed to confirmed/suspected hardware issues
 - 58.7% were GPU-related: including NVLink failures and HBM3 memory errors

Source: <https://aws.amazon.com/blogs/machine-learning/introducing-amazon-eks-support-in-amazon-sagemaker-hyperpod/>

Business challenges

- 1 Hardware failures disrupt and extend training cycles
- 2 Static compute allocation leads to underutilization
- 3 High-performance compute is scarce and difficult to scale
- 4 Cluster customization adds complexity and slows development
- 5 Lack of prescriptive observability of cluster and model performance



All of this leads to:

Inefficiency, slower time-to-train, and ultimately...
higher costs

Amazon SageMaker HyperPod



What is Amazon SageMaker HyperPod?

A fully-resilient cluster designed for FM training efficiency at scale, improving total cost of ownership (TCO) for FM builders. Develop and train FMs continuously for weeks and months.



Business challenges addressed

Improved efficiency

Tools for maximizing utilization, advanced observability, and seamless cluster customization

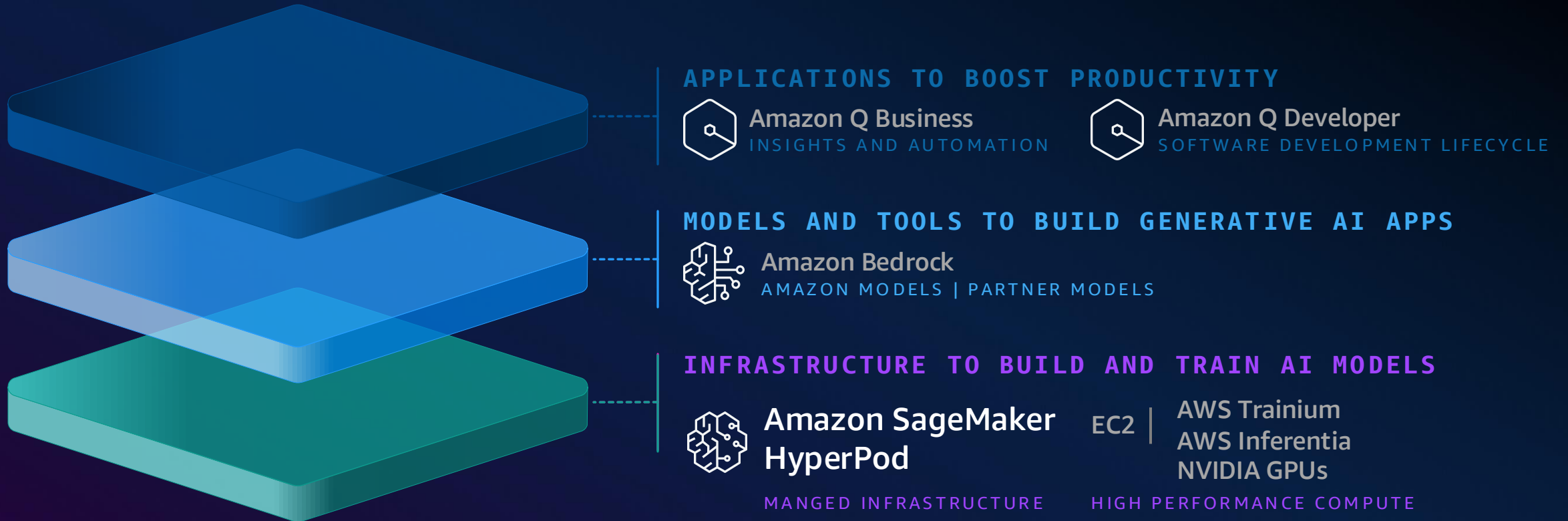
Reduced time-to-train

Resilience features and SageMaker libraries can reduce time to train by up to 40%

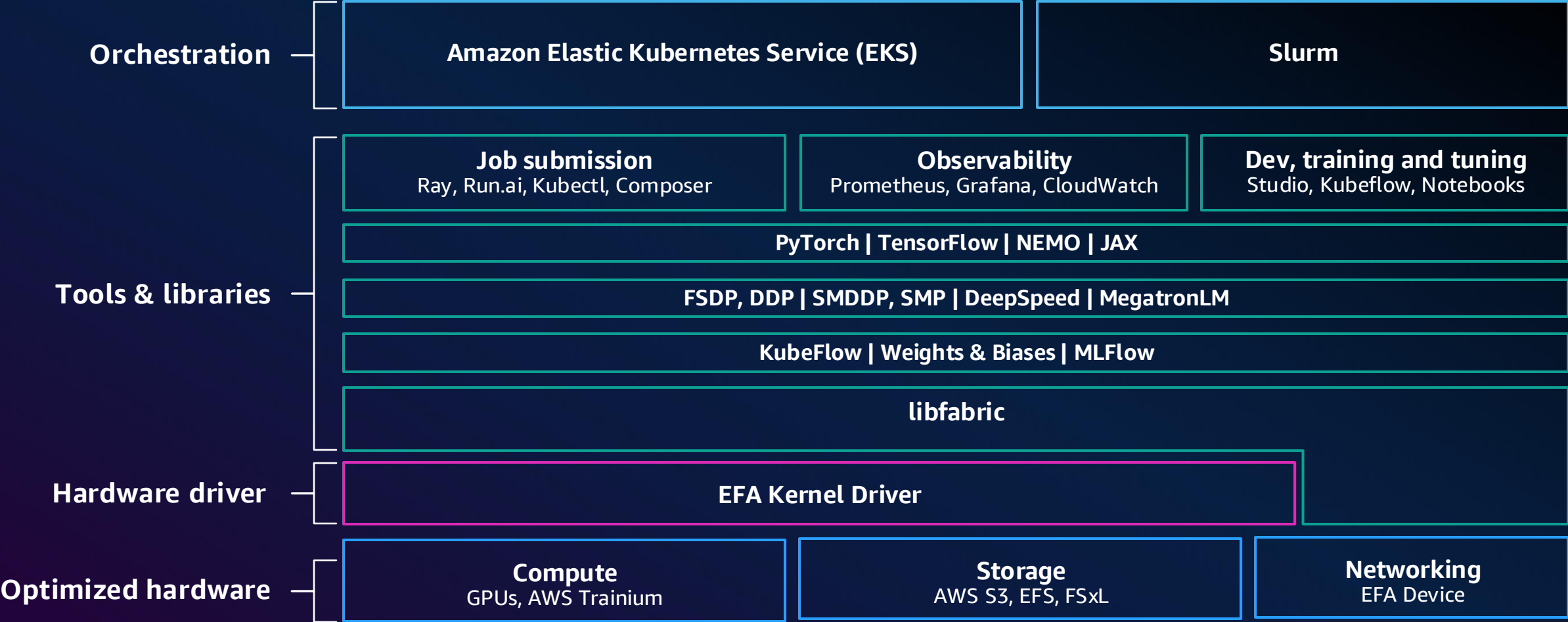
Lower costs

Less time invested in hardware maintenance and more efficient cluster engagement reduces FM training TCO

Amazon SageMaker HyperPod within the AWS GenAI Stack



Amazon SageMaker HyperPod Environment



Recipes

How it works

- 1 Select a recipe from popular publicly-available FMs (DeepSeek, Llama, etc.)

- 2 Specify training and validation data directories

- 3 Run the recipe on SageMaker HyperPod or SageMaker Training Jobs


Business value

- Reduce FM training setup time and get started in minutes vs. days or weeks
- Pre-tested training stack configurations for fine-tuning and pre-training publicly available FMs
- Easily switch between GPU and Trainium instances
- Open-source and customizable, with a growing list of 90+ recipes



Flexible training plans

Business value

- Easily secure future capacity for core training, inference, or bursting workloads
- Self-service and simple
- Reserves capacity for up to 6 months
- Lower cost (up to 68% savings vs. on-demand pricing)

Training plan requirements

Target

☐ Training job

☒ HyperPod cluster

Instance attributes

Instance type	Instance count
ml.trn2.48xlarge	8

Date settings to search for an available plan

You can enter a start date in as little as 30 minutes, up to 8 weeks in advance.

Start date	End date - optional
2025/04/16	YYYY/MM/DD

Duration (Days)

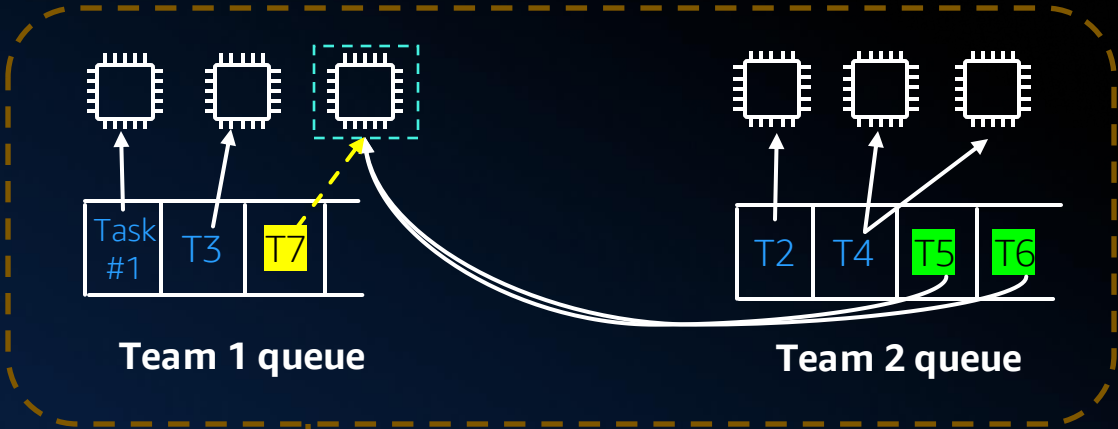
[Find training plan](#)

How it works

Specify...

- 1
 - Instance type (p5, p5e, p5en, trn1, trn2, p4d)
 - Number of instances (1-64)
 - Duration of plan (1-182 days)
 - Start date (within 8 weeks)
- 2
 - Review available segments
 - Confirm selections
- 3
 - Use the plan when creating HyperPod clusters

Task governance



Problem: Static compute allocation leads to underutilization

Tasks execute inefficiently in flat queues

Teams compete for scarce compute resources

Un-prescriptive accelerator utilization monitoring

Solution: Dynamic utilization of idle compute resources

Advanced task prioritization and preemption policies

Adaptive allocation across large and diverse teams

Deep insights into resource utilization, task health, and allocations

Business value

Up to 40% improved cluster utilization

Lower costs

Reduced task wait-times

Higher team efficiency

Observability and experimentation tools



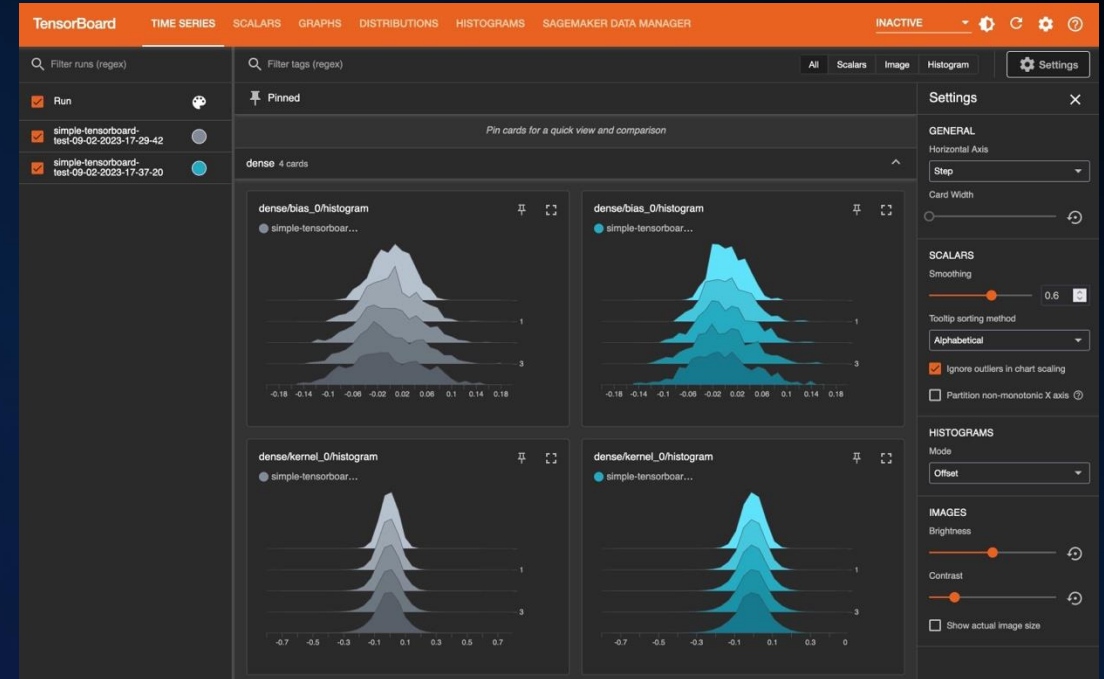
Managed TensorBoard on SageMaker provides visualization of model architecture to identify and remediate convergence issues



Amazon CloudWatch Container Insights provides deeper understanding of cluster performance

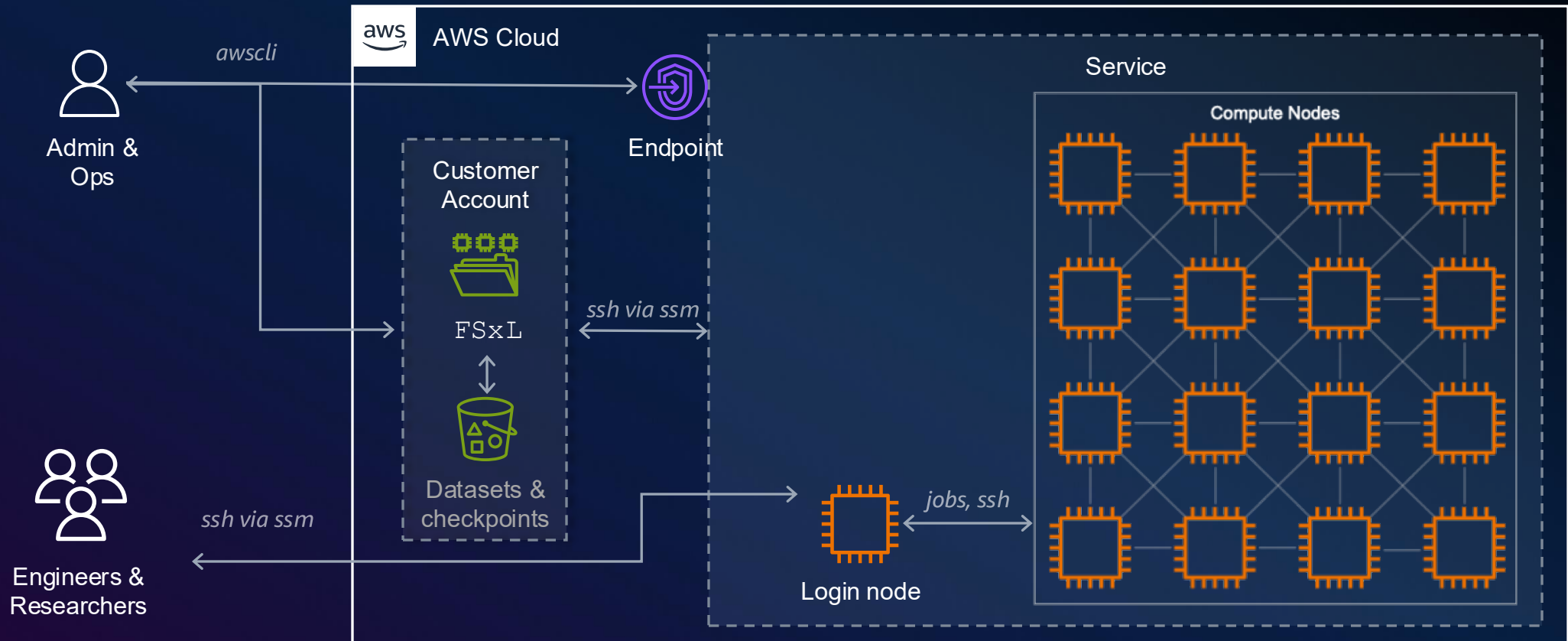


Managed MLflow on SageMaker enables efficient model experimentations at scale

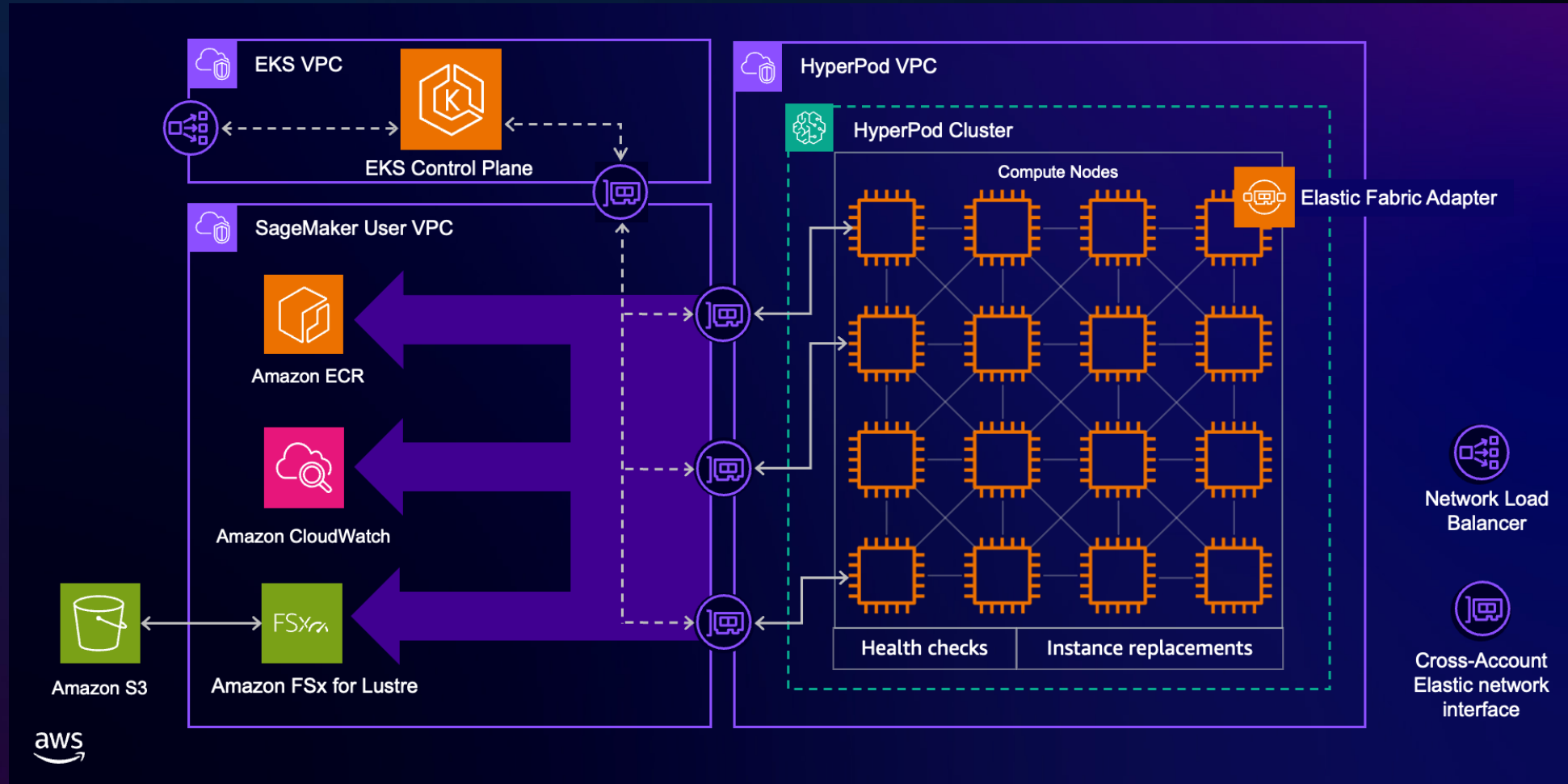


Orchestrating Foundation Model Training at scale

Orchestration: Slurm



Orchestration: Amazon EKS



Slurm or Amazon EKS? What's right for your team?

Feature	Slurm	Amazon EKS
Control plane ownership	Controller node is a part of HyperPod cluster	Control plane is owned by EKS
Login to cluster	Customers need to login in to head node or login node to run Slurm commands	Customers can run kubectl from remote machines
Cluster metrics observability	Customers can setup observability stack at application level, but it required extra effort.	Customers can use CloudWatch container insights as a first class observability feature
When resiliency runs	Instance replacement is triggered only when `--auto-resume=1` is specified to srun command, and when hardware failed.	Health monitoring runs in the background, and instance reboot/replacement happens anytime hardware issue is detected.
Deep health checks	Deep health check doesn't exist	Customers can enable deep health checks
Instance reboot	Customers can reboot by hitting "sudo reboot" command on the instance, but there is no way to reboot unresponsive instances.	Customers can reboot instances by setting node label and it works even if the instance is unresponsive.
Task governance	You can use Slurm's QoS feature	HyperPod task governance is supported for scheduling/priority/dynamic compute resource allocation
HyperPod CLI	Not supported	Supported

Is Amazon SageMaker HyperPod right for your team?

Key factors to consider:

- **Model size and complexity**
- **Training frequency and duration**
- **Budget control**
- **Team workflows**
- **Infrastructure persistence**

Customer stories



Resilient computing with Amazon SageMaker HyperPod

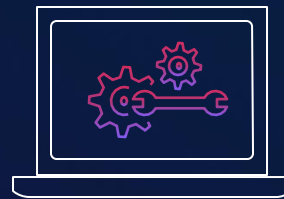


Stability AI

- Leading open-source GenAI company
- **Challenge:** Losing hundreds of hours to hardware failures



HyperPod automatically replaces faulty instances once detected



Workloads are automatically resumed – **saving Stability AI over 50% in training time and cost**



Performant computing with SageMaker HyperPod



Perplexity AI

- First conversational Q&A engine
- **Challenge:** low throughput of LLM training experiments



SageMaker distributed training libraries provide an optimized backend for deep learning models in TensorFlow, PyTorch



Up to 20% faster and cheaper than NCCL and MPI-based solutions – **Unlocking a 2x increase in experiment throughput for Perplexity AI**

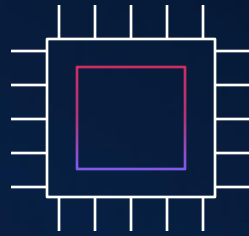


Customizable computing with Amazon SageMaker HyperPod

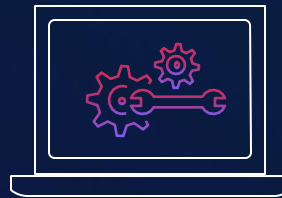


Hugging Face

- Open source machine learning platform and community
- **Challenge:** Research requires control over all layers of tech stack



Customize HyperPod clusters with frameworks, containers, debugging tools



Hugging Face's science teams has the **freedom to innovate quickly**

Amazon SageMaker HyperPod Resources



Amazon SageMaker HyperPod
Overview



Amazon SageMaker HyperPod
Developer Guide



Amazon SageMaker HyperPod
Blog Post



Amazon SageMaker HyperPod
Workshop



AWSome Distributed Training

skillbuilder.aws 

Create a free account
on AWS Skill Builder to
gain in-demand skills

Thank you!

Aastha Varma

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